

8.6 Consider the wage equation

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i \quad (XR8.6a)$$

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

- a. We are curious whether holding education, experience, and $METRO$ constant, there is the same amount of random variation in wages for males and females. Suppose $\text{var}(e_i | x_i, FEMALE = 0) = \sigma_M^2$ and $\text{var}(e_i | x_i, FEMALE = 1) = \sigma_F^2$. We specifically wish to test the null hypothesis $\sigma_M^2 = \sigma_F^2$ against $\sigma_M^2 \neq \sigma_F^2$. Using 577 observations on males, we obtain the sum of squared OLS residuals, $SSE_M = 97161.9174$. The regression using data on females yields $\hat{\sigma}_F^2 = 12.024$. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

a.

★ 已知 information :

$$\text{Male} : N = 577, SSE_M = 97161.9174$$

$$\text{Female} : N = 100 - 577 = 423, \hat{\sigma}_F^2 = 12.024$$

★ Calculate SSE_F

$$\hat{\sigma}_F^2 = \frac{SSE_F}{N - K} \Rightarrow 12.024 = \frac{SSE_F}{423 - 4} \Rightarrow SSE_F = 60,577.59$$

★ F statistic

$$\hat{\sigma}_M^2 = \frac{SSE_M}{N - K} = \frac{97161.9174}{577 - 4} = 169.57$$

$$\hat{\sigma}_F^2 = 144.58$$

$$F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} = \frac{169.57}{144.58} = 1.173 \sim F_{573, 419}$$

★

$$H_0 : \sigma_M^2 = \sigma_F^2, H_1 : \sigma_M^2 \neq \sigma_F^2$$

$$F_{(1 - 0.025, 573, 419)} \approx 1.1968$$

$$F_{(0.025, 573, 419)} \approx 0.8377$$

$$\text{Rejection region} : \{ F > 1.1968 \text{ or } F < 0.8377 \}$$

$$F^* = 1.173 \text{ doesn't fall in RR, we won't reject } H_0.$$

$$\text{Thus, there is not enough evidence to prove that } \sigma_M^2 \neq \sigma_F^2$$

- b. We hypothesize that **married individuals**, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages. Suppose $\text{var}(e_i|x_i, \text{MARRIED} = 0) = \sigma_{\text{SINGLE}}^2$ and $\text{var}(e_i|x_i, \text{MARRIED} = 1) = \sigma_{\text{MARRIED}}^2$. Specify the null hypothesis $\sigma_{\text{SINGLE}}^2 = \sigma_{\text{MARRIED}}^2$ versus the alternative hypothesis $\sigma_{\text{MARRIED}}^2 > \sigma_{\text{SINGLE}}^2$. We add **FEMALE** to the wage equation as an explanatory variable, so that

$$\text{WAGE}_i = \beta_1 + \beta_2 \text{EDUC}_i + \beta_3 \text{EXPER}_i + \beta_4 \text{METRO}_i + \beta_5 \text{FEMALE} + e_i \quad (\text{XR8.6b})$$

Using **$N = 400$** observations on single individuals, OLS estimation of (XR8.6b) yields a sum of squared residuals is **56231.0382**. For the **600** married individuals, the sum of squared errors is **100,703.0471**. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

b.

★ 已知 information :

$$\text{Single} : N = 400, SSE_S = 56231.0382$$

$$\text{Married} : N = 600, SSE_M = 100703.0471$$

★ Calculate σ^2

$$\sigma_S^2 = \frac{SSE}{N-K} \Rightarrow \sigma_S^2 = \frac{56231.0382}{400-5} = 142.36$$

$$\sigma_M^2 = \frac{100703.0471}{600-5} = 169.25$$

★ F statistic

$$F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_S^2} = \frac{169.25}{142.36} = 1.189 \sim F_{595, 395}$$

★

$$H_0: \sigma_M^2 = \sigma_S^2, H_1: \sigma_M^2 > \sigma_S^2$$

$$F_{(0.05, 595, 395)} \approx 1.1647$$

$$\text{Rejection region} : \{ F > 1.1647 \}$$

$F = 1.189$ falls in RR, we reject H_0 .

Thus, we conclude that $\sigma_M^2 \neq \sigma_S^2$.

- c. Following the regression in part (b), we carry out the **NR² test** using the right-hand-side variables in (XR8.6b) as candidates related to the heteroskedasticity. The value of this statistic is **59.03**. What do we conclude about heteroskedasticity, at the 5% level? Does this provide evidence about the issue discussed in part (b), whether the error variation is different for married and unmarried individuals? Explain.

c.

★ **H₀ v.s. H₁**

$$H_0 : \alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = \alpha_5 = 0$$

H₁ : Not all the α_i in H₀ are zero

★ **X² Statistic**

$$NR^2 = 59.03 \sim \chi^2(4)$$

$$\chi^2_{(1-0.05, 4)} = 9.488$$

$$\chi^2 = 59.03 > \chi^2_{(0.95, 4)}, \text{ we reject } H_0.$$

There exists heteroskedasticity problem. However, the NR² test only tells us that the error variance is related to some explanatory variables, but cannot directly point out that it is caused by the specific variable MARRIED.

- d. Following the regression in part (b) we carry out the **White test** for heteroskedasticity. The value of the test statistic is **78.82**. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?

d.

★ **Variables**

$$X : EDUC, EXPER, METRO, FEMALE \Rightarrow 4$$

$$X^2 : EDUC^2, EXPER^2, METRO^2, FEMALE^2 \Rightarrow 4$$

$$\text{cross product} : \frac{4 \times 3}{2} \Rightarrow 6$$

★ **test statistic**

H₀ : homoskedasticity v.s. H₁ : heteroskedasticity

$$\chi^2_{(1-0.05, 14)} = 23.685$$

$$\chi^2 = 78.82 > \chi^2_{(1-0.05, 14)} = 23.685 \Rightarrow \text{We reject } H_0.$$

We can conclude that there is heteroskedasticity problem.

- e. The OLS fitted model from part (b), with usual and robust standard errors, is

$$\widehat{WAGE} = -17.77 + 2.50EDUC + 0.23EXPER + 3.23METRO - 4.20FEMALE$$

(se)	(2.36)	(0.14)	(0.031)	(1.05)	(0.81)
(robse)	(2.50)	(0.16)	(0.029)	(0.84)	(0.80)

For which coefficients have interval estimates gotten narrower? For which coefficients have interval estimates gotten wider? Is there an inconsistency in the results?

e.

narrower interval estimates : EXPER, METRO, FEMALE

wider interval estimates : EDUC

This is not contradictory. When heteroskedasticity exists, some coefficients of the general SE in OLS will be overestimated and some underestimated. After robust SE correction, the SE of some coefficients will decrease (originally overestimated) and some will increase (originally underestimated).

- f. If we add *MARRIED* to the model in part (b), we find that its *t*-value using a White heteroskedasticity robust standard error is about 1.0. Does this conflict with, or is it compatible with, the result in (b) concerning heteroskedasticity? Explain.

f.

There is no contradiction.

GA test : to check whether there exists heteroskedasticity.

t test : to check whether the average effect of *MARRIED* is significant.

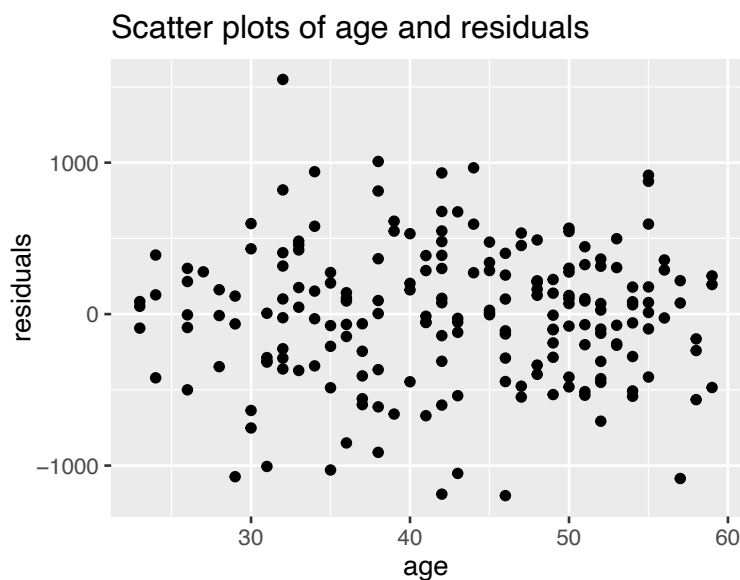
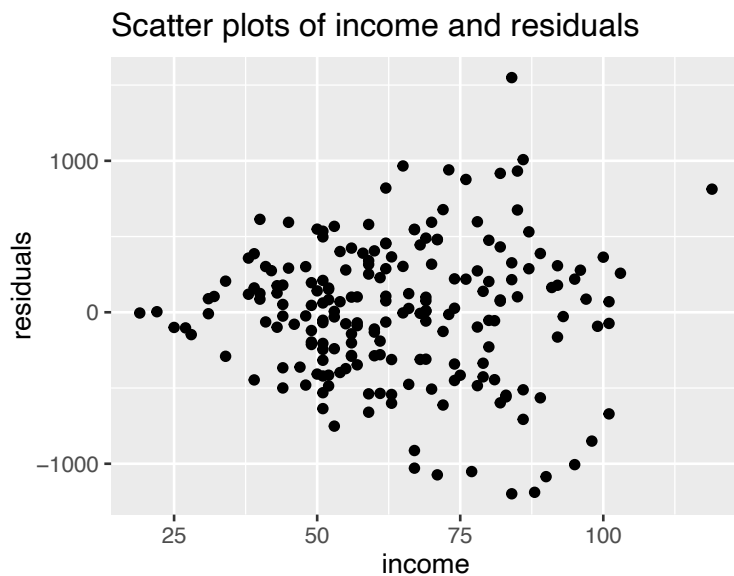
8.16

a.

- 95% interval estimate for the effect of one more child on miles traveled: [-135.3298, -28.3230]

	2.5 %	97.5 %
(Intercept)	-726.36871	-56.72731
income	10.65097	17.75169
age	8.33086	23.15099
kids	-135.32981	-28.32302

b.



- The plot of residuals versus **income** shows that residual variance increases with income, which strongly suggests **heteroskedasticity**. In contrast, the spread of residuals remains **relatively constant** across different ages, indicating no obvious issue with age-related error variance.

c.

- $H_0: \sigma^2_{low} = \sigma^2_{high}$
- $H_1: \sigma^2_{low} \neq \sigma^2_{high}$
- GQ Statistic = 3.1041.
- Given the critical values are 0.6534 and 1.5304, since $F = 3.1041 > 1.5304$, we reject the null hypothesis H_0 .

```
> flc <- qf(alpha/2, df1, df2)
> fuc <- qf(1-alpha/2, df1, df2)
> cat(flc, fuc, qq_fstat)
0.6534355 1.530373 3.104061
```

d.

- 95% interval estimate for the effect of one more child on miles traveled: [-139.3229, -24.3298]
- The 95% confidence interval for the effect of kids constructed using robust standard error is **wider** than the conventional OLS method in part a.

e.

- 95% interval estimate for the effect of kids constructed using conventional GLS: [-119.8945, -33.7180]

```
> print(ci_conv_gls)
[1] -119.89450 -33.71808
```
- 95% interval estimate for the effect of kids constructed using robust GLS standard error: [-121.4133, -32.1991]

```
> print(ci_robust_gls)
[1] -121.41339 -32.19919
```
- The interval estimates obtained from conventional and robust GLS are narrower than those from part a and d. These narrower intervals represent more precise estimates of the effect of kids.